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Early activation of mTORC1 signalling in response to mechanical overload is independent of phosphoinositide 3-kinase/Akt signalling

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Non-technical summary Hypertrophy of skeletal muscle in response to resistance exercise is associated with significantly elevated rates of protein synthesis. The protein kinase mTORC1 has been shown to be a key signalling hub through which different anabolic factors (i.e. growth factors, nutrients and mechanical strain) contribute to the regulation of protein synthesis. In this study, we use an *in vivo* model of muscle hypertrophy to delineate the contribution of different input pathways regulating mTORC1. We found that the insulin/insulin like growth factor 1 pathway is not necessary for early activation of mTORC1 signalling but this probably occurs through activation of the ERK/TSC2 pathway. Knowledge of the key upstream pathways that modulate mTORC1 activity *in vivo* will provide the necessary foundation for the development of new therapeutic strategies for the maintenance of skeletal muscle mass.

Abstract The mammalian target of rapamycin complex 1 (mTORC1) functions as a central integrator of a wide range of signals that modulate protein metabolism and cell growth. However, the contributions of individual pathways regulating mTORC1 activity in skeletal muscle are poorly defined. The purpose of this study was to determine the regulatory mechanisms that contribute to mTORC1 activation during mechanical overload-induced skeletal muscle hypertrophy. Consistent with previous studies, mechanical overload induced progressive hypertrophy of the plantaris muscle which was associated with significant increases in total RNA content and protein metabolism. mTORC1 was activated after a single day of overload as indicated by a significant increase in S6K1 phosphorylation at T389 and T421/S424. In contrast, Akt activity, as assessed by Akt phosphorylation status (T308 and S473), phosphorylation of direct downstream targets (glycogen synthase kinase 3 β , proline-rich Akt substrate 40 kDa and tuberous sclerosis 2 (TSC2)) and a kinase assay, was not significantly increased until 2–3 days of overload. Inhibition of phosphoinositide 3-kinase (PI3K) activity by wortmannin was sufficient to block insulin-dependent signalling but did not prevent the early activation of mTORC1 in response to overload. We identified that the mitogen-activated protein kinase kinase (MEK)/extracellular signal-regulated kinase (ERK)-dependent pathway was activated at day 1 after overload. In addition, a target of MEK/ERK signalling, phosphorylation of TSC2 at S664, was also increased at this early time point. These observations demonstrate that *in vivo*, mTORC1 activation at the early phase of mechanical overload in skeletal muscle occurs independently of PI3K/Akt signalling and provide evidence that the MEK/ERK pathway may contribute to mTORC1 activation through phosphorylation of TSC2.

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Abbreviations DAPI, 4',6-diamidino-2-phenylindole; ERK, extracellular signal-regulated kinase; GSK3- β , glycogen synthase kinase 3 β ; IGF-1, insulin-like growth factor-1; MAPK, mitogen-activated protein kinase; MEK, mitogen-activated protein kinase kinase; mTOR, mammalian target of rapamycin; mTORC1, mammalian target of rapamycin complex 1; mTORC2, mammalian target of rapamycin complex 2; PI3K, phosphoinositide 3-kinase; PRAS40, proline-rich Akt substrate 40 kDa; rpS6, ribosomal protein S6; RSK, ribosomal S6 kinase; S6K1, S6-kinase 1; TSC, tuberous sclerosis.

Introduction

The maintenance of skeletal muscle mass throughout life is important for long-term health and quality of life. Skeletal muscle mass is generally determined by the net balance between protein synthesis and protein degradation. Following an increase in workload, the rate of protein synthesis in skeletal muscle is enhanced relative to protein degradation rates such that there is a net increase in cellular protein leading to skeletal muscle hypertrophy (Miyazaki & Esser, 2009a). To date, many studies have shown that mammalian target of rapamycin (mTOR) plays a critical role in regulating the rate of protein synthesis and subsequent hypertrophy in skeletal muscle (Bodine *et al.* 2001; Rommel *et al.* 2001; Hornberger *et al.* 2004; Nader *et al.* 2005). mTOR is a serine/threonine kinase of the phosphatidylinositol kinase-related kinase family that is highly conserved from yeast to mammals (Jacinto & Hall, 2003). The function of mTOR is largely determined by its association with two different multi-protein complexes, mTOR complex 1 (mTORC1) and mTOR complex 2 (mTORC2). mTORC1 is a rapamycin-sensitive protein complex known to be composed of five proteins: mTOR, regulatory-associated protein of mTOR (Raptor), G protein β -subunit-like (G β /also known as mLST8), proline-rich Akt substrate 40 kDa (PRAS40) and DEP-domain-containing mTOR-interacting protein (Deptor). mTORC2 is rapamycin-insensitive and is composed of at least six different proteins, some of which overlap with mTORC1: mTOR, rapamycin-insensitive companion of mTOR (Rictor), stress-activated-protein-kinase-interacting protein 1, protein observed with Rictor-1, G β 1 and Deptor (see review by Laplante & Sabatini, 2009).

In skeletal muscle cells, mTORC1 functions as a central integrator of a wide range of signals that promote muscle hypertrophy by increasing protein synthesis rates (Miyazaki & Esser, 2009a). Specifically, mTORC1 regulates protein synthesis rates by phosphorylating the downstream effectors, eukaryotic initiation factor 4E-binding protein 1 and the p70 ribosomal S6 kinase 1 (S6K1). Currently, the most well-defined signalling mechanism regulating mTORC1 activity in skeletal muscle is the insulin/insulin-like growth factor-1 (IGF-1)-dependent pathway (Bodine *et al.* 2001; Rommel *et al.* 2001). Growth factor (e.g. IGF-1 and/or insulin) stimulation of

muscle leads to activation of phosphoinositide 3-kinase (PI3K) and its downstream effector Akt (also referred to as protein kinase B), triggering multiple downstream signalling events including mTORC1 activation (Frost & Lang, 2007; Miyazaki *et al.* 2010). The ability of IGF-1 to act as an anabolic agent in an autocrine/paracrine fashion in skeletal muscle has been clearly demonstrated in both the *in vitro* cell culture model and the *in vivo* animal model. The administration of exogenous IGF-1 to cultured myotubes (Rommel *et al.* 2001; Vyas *et al.* 2002) or via localized infusion into skeletal muscle (Adams & McCue, 1998) results in a robust increase in muscle cell size and total protein content. In addition, over-expression of a constitutive-active Akt in skeletal muscle resulted in a pronounced hypertrophic phenotype (Lai *et al.* 2004). Collectively, these studies show that the addition of exogenous IGF-1 and the subsequent activation of PI3K/Akt-dependent signalling can effectively induce skeletal muscle hypertrophy.

Recently, the necessity of IGF-1 for muscle hypertrophy was directly challenged by Spangenburg and colleagues (Spangenburg *et al.* 2008). In mice in which IGF-1 signalling was blocked by over-expression of a dominant-negative IGF-1 receptor, skeletal muscle hypertrophied to the same degree as in wild-type mice in response to mechanical overload. Importantly, the mTORC1-dependent signalling pathway, which is considered to be necessary for skeletal muscle hypertrophy, was equally activated in skeletal muscle via mechanical overload in both wild-type and the dominant-negative IGF-1 receptor transgenic mice. These observations suggest that activation of IGF-1-dependent signalling is not an absolute requirement for the induction of skeletal muscle hypertrophy in response to increased mechanical loading *in vivo*. Furthermore, these findings are consistent with earlier reports showing that activation of mTORC1 following mechanical strain either *in vitro* (C2C12 model) (Hornberger *et al.* 2004) or *ex vivo* (O'Neil *et al.* 2009) was independent of PI3K/Akt regulation.

Despite mTORC1 being a crucial regulator of protein synthesis and hypertrophy in skeletal muscle, the regulatory mechanisms which govern mTORC1 activity *in vivo* remain unresolved. The purpose of this study was to identify the signalling pathways contributing to mTOR activation during the initial stage of mechanical overload-induced skeletal muscle hypertrophy *in vivo*.

Methods

Materials

Insulin and protease inhibitor cocktail for mammalian tissues were from Sigma-Aldrich (St Louis, MO, USA). Wortmannin and rapamycin were from Calbiochem (San Diego, CA, USA). Protein assay dye reagent concentrate was from Bio-Rad Laboratories (Hercules, CA, USA). ECL and ECL Plus solutions were obtained from GE Healthcare Bioscience (Piscataway, NJ, USA). Akt activity assay kit was from Abcam (Cambridge, MA, USA). TRIzol reagent was from Invitrogen (Carlsbad, CA, USA). TURBO DNA-free was from Applied Biosystems (Foster City, CA, USA). Protein A Plus agarose was from Thermo Fisher Scientific (Rockford, IL, USA). VECTASTAIN ABC Kit for rabbit IgG and VECTASHIELD mounting medium with 4',6-diamidino-2-phenylindole (DAPI) were from Vector Laboratories (Burlingame, CA, USA). L-[2,3,4,5,6-³H]Phenylalanine was from Amersham Life Science (Arlington Heights, IL, USA). **Antibodies:** phospho-S6K1 (T389), phospho-S6K1 (T421/S424), Akt, phospho-Akt (T308), phospho-Akt (Ser473), PRAS40, phospho-PRAS40 (T246), phospho-GSK3- β (S9), TSC1/hamartin, phospho-TSC2 (S939), phospho-TSC2 (T1462), phos-MEK1/2 (S217/221), MEK1/2, phos-ERK1/2 (T202/204), ERK1/2, phos-rpS6 (S235/236), phos-rpS6 (S240/244), rpS6, phos-RSK (T359/S363), phos-RSK (S380) and RSK1/2/3 were from Cell Signaling Technology (Danvers, MA, USA). GSK3- β and CD31 were from BD Biosciences (San Diego, CA, USA). phospho-TSC2 used for immunohistochemistry (S664) was from Biolegend (San Diego, CA, USA). TSC2/tuberin (C-20) and S6K1 (C-18) were from Santa Cruz Biotechnology (Santa Cruz, CA, USA). GAPDH was from Abcam. Peroxidase-labelled anti-rabbit IgG and anti-mouse IgG secondary antibodies were from Vector Laboratories. Texas Red-labelled goat anti-rabbit IgG and Alexa Fluor 488-conjugated goat anti-rat IgG were from Molecular Probes (Eugene, OR, USA). The phospho-TSC2 (S664) antibody used for Western blotting was kindly provided by Dr Pier Paolo Pandolfi, Beth Israel Deaconess Medical Center, Harvard Medical School.

Animal care and use

All experimental procedures performed in this study were approved by the University of Kentucky Institutional Animal Care and Use Committee. Animals were housed in temperature- and humidity-controlled holding facilities on a 14:10 h light:dark cycle and had access to food and water *ad libitum*. Male C57BL/6J mice (The Jackson Laboratory), 12–14 weeks of age, were used in this study. The bilateral synergist ablation model was used to induce hypertrophy of the plantaris muscle as described previously (Miyazaki *et al.* 2004; McCarthy & Esser, 2007).

This *in vivo* model of hypertrophy was originally described by Dr A. L. Goldberg (Goldberg, 1967) and induces hypertrophic growth of the plantaris muscle through mechanical overload resulting from the surgical removal of the synergist muscles i.e. the gastrocnemius and soleus muscles. Insulin (0.75 U kg^{-1}), wortmannin (3 mg kg^{-1}) and rapamycin (1.5 mg kg^{-1}) were administered by intraperitoneal injection. DMSO was used as a solvent for wortmannin and rapamycin. At the end of each experimental period, mice were anaesthetized with an intraperitoneal injection of ketamine (100 mg kg^{-1}) and xylazine (10 mg kg^{-1}), and the muscles were excised, weighed, quickly frozen in liquid nitrogen, and stored at -80°C . Upon the completion of experimental treatments, mice were killed by cervical dislocation under anaesthesia.

In vitro measurement of protein synthesis/degradation rates

Rates of protein synthesis and degradation were measured *in vitro* as previously described (Fedele *et al.* 2001; Hornberger *et al.* 2004). Rates of protein synthesis, expressed as nanomoles of phenylalanine incorporated per milligram of protein per hour, were measured by the incorporation of radioactive phenylalanine from the incubation medium into the plantaris muscle. The plantaris muscles were excised intact and immediately placed in Krebs–Henseleit bicarbonate (KHB) buffer at optimal resting length. Muscles were then transferred into 3 ml of fresh KHB buffer supplemented with 5 mCi ml^{-1} of L-[2,3,4,5,6-³H]phenylalanine and incubated for 2 h. During the incubation period, the KHB buffer was oxygenated with 95% oxygen–5% carbon dioxide gas and maintained at 37°C . Following the incubation, muscles were homogenized in 10% trichloroacetic acid (TCA) and centrifuged at $10,621 \text{ g}$ for 15 min at 4°C . The pellet was washed with 10% TCA and re-suspended in 1 N sodium hydroxide. Aliquots were assayed for total protein using a Lowry protein assay and measured for radioactivity by liquid scintillation counting (Beckman model LS-6500) with appropriate correction for quench.

Rates of protein degradation, expressed as nanomoles of tyrosine released per milligram of protein per hour, were measured by the accumulation of tyrosine in the incubation medium. Tyrosine in the medium was measured fluorometrically as described previously (Fedele *et al.* 2001). Total protein degradation was determined as the sum of tyrosine accumulation in the incubation medium plus the amount of tyrosine equivalents incorporated into the muscle during the 2 h incubation period. The amount of tyrosine incorporated into muscle was determined by multiplying the amount of phenylalanine incorporated by 0.77, which is the molar ratio of tyrosine to phenylalanine in mixed skeletal muscle proteins.

RNA isolation, protein extraction and Western blotting

Total RNA was prepared from frozen tissue samples using TRIzol reagent according to the manufacturer's directions. RNA samples were treated with TURBO DNA-free to remove genomic DNA contamination. Isolated RNA was quantified by spectrophotometry ($\lambda = 260$ nm). To prepare total protein lysate, frozen muscle samples were homogenized in ice-cold RIPA buffer (1% NP-40, 0.5% sodium deoxycholate, 0.1% SDS, 50 mM sodium chloride, 20 mM Tris-HCl (pH 7.6), 1 mM phenylmethylsulfonyl fluoride (PMSF), 5 mM benzamidine, 1 mM EDTA, 5 mM *N*-ethylmaleimide, 50 mM sodium fluoride, 25 mM β -glycerophosphate, 1 mM sodium orthovanadate, $10 \mu\text{l ml}^{-1}$ protease inhibitor cocktail). Homogenates were then centrifuged at 17,860 *g* for 10 min at 4°C and the supernatant collected for analysis. Protein concentration was determined by the Bradford method. Protein extracts were run on SDS-PAGE gels and transferred to polyvinylidene difluoride membranes. Membranes were blocked in 5% non-fat dry milk in Tris-buffered saline mixed with 0.1% Tween 20 and then incubated with dilutions of each primary antibody. A horseradish peroxidase-conjugated anti-rabbit IgG was used as secondary antibody with bound antibody complexes visualized using ECL or ECL Plus Western blotting detection reagents followed by exposure to X-ray film. The X-ray film images were scanned and band intensity for a protein of interest was quantified using Image J software (NIH, USA).

Immunoprecipitation assay

To study the functional interactions between tuberous sclerosis 1 (TSC1) and TSC2, co-immunoprecipitation assays were performed as described previously (Miyazaki & Esser, 2009b). NP-40-based buffer (1% NP-40, 10 mM Tris-HCl (pH 7.6), 100 mM sodium chloride, 50 mM sodium fluoride, 2 mM EDTA, 1 mM PMSF, $10 \mu\text{l ml}^{-1}$ protease inhibitor cocktail) was used to produce total cell lysate. For each antibody, immunocomplexes from 0.5 mg of lysate were pulled down using immobilized protein A. Immunocomplexes were washed three times with NP-40-based buffer and then washed once with wash buffer (50 mM Hepes (pH 7.5), 40 mM sodium chloride and 2 mM EDTA). Precipitated protein samples were then subjected to SDS-PAGE as described above.

In vitro Akt kinase assay

Akt kinase activity was determined by Akt Activity Assay Kit according to the manufacturer's directions. Briefly, skeletal muscle samples were homogenized in

ice-cold kinase extraction buffer. Homogenates were then centrifuged at 17,860 *g* for 10 min at 4°C and the supernatant collected for analysis. Cell lysates (400 μg total protein) were immunoprecipitated with Akt specific antibody, pulled down using protein A sepharose and then immunocomplexes were washed twice with kinase extraction buffer and once with kinase assay buffer. Fifty microlitres of kinase assay buffer and 2 μl of GSK3- α protein/ATP mixture were added to the washed protein A beads and then incubated at 30°C for 4 h. Following the incubation, phosphorylation states of GSK3- α were determined by Western blotting as described above.

Immunohistochemistry

Plantaris muscles were prepared for immunohistochemistry by first being frozen in liquid nitrogen-cooled isopentane. Cross-sections (10 μm) were cut from the midbelly portion of the muscle in a cryostat (Microme HM525) and stored at -80°C until analysis. Frozen sections were air-dried and fixed with 4% paraformaldehyde in phosphate-buffered saline (PBS) (pH 7.5). Sections were then washed with PBS-T (Tween 20 at 0.05% in PBS) and permeabilized with 0.1% Triton X-100 in PBS. For the immunofluorescent staining of phosphorylated-rpS6 and CD31, sections were blocked with 1% bovine serum albumin. Rabbit anti-phos-rpS6 (S235/236) and Texas Red-conjugated goat anti-rabbit antibodies were used for detecting rpS6 phosphorylation. Rat anti-mouse CD31 and Alexa Fluor 488-conjugated goat anti-rat antibodies were used for endothelial cell marker CD31 localization. VECTASHIELD mounting medium with DAPI was used for nuclei counterstaining. For the immunoperoxidase staining of phosphorylated-TSC2 at S664, VECTASTAIN ABC kit was used according to the manufacturer's directions. Briefly, fixed and permeabilized sections were blocked with goat serum. Sections were then incubated with phospho-TSC2 (S664) antibody overnight. The next day, sections were washed with PBS-T and incubated first with a biotinylated secondary antibody and then with VECTASTAIN ABC reagent; colour development was followed under the microscope using diaminobenzidine. All images were captured using the Nikon ECLIPSE E600 system and SPOT Advanced Software (SPOT Imaging Solutions, Sterling Heights, MI, USA).

Statistical analysis

All values are reported as means $\pm$ SEM. Multi-group comparisons were performed by one-way analysis of variance followed by Tukey's *post hoc* test. For all comparisons, the level of statistical significance was set at $P < 0.05$.

Results

Muscle hypertrophy associated with enhanced rate of protein synthesis

The changes in plantaris muscle weight and total RNA content were determined from 1 to 10 days of mechanical overload induced by synergist ablation. As shown in Fig. 1A, plantaris muscle weight was significantly increased after 1 day of overload (OV-1) and continued to progressively increase at each time point over the next 10 days, reaching a 2.1-fold increase by OV-10. Similarly, total RNA was significantly increased by 2.1-fold after 5 days of overload (OV-5) and remained elevated throughout the remainder of the time period (Fig. 1B). Associated with the change in muscle weight and total RNA content, protein turnover, as assessed by measuring rates of protein synthesis and degradation under *in vitro*

conditions, was significantly increased at OV-7 with a 14.2-fold increase in protein synthesis and 1.5-fold increase in rates of protein degradation (Fig. 1C and D).

Early activation of mTORC1 is independent of Akt activation

Given the increase in the rate of protein synthesis in response to mechanical overload, we next wanted to define the time point at which Akt and mTORC1-dependent signalling was activated. S6K1, which phosphorylates ribosomal protein S6 (rpS6), is the one of most well-characterized downstream targets of mTORC1. Particularly, phosphorylation of S6K1 at T389 is often used as a functional readout of mTORC1 activity, since this residue is specifically phosphorylated by mTORC1 under

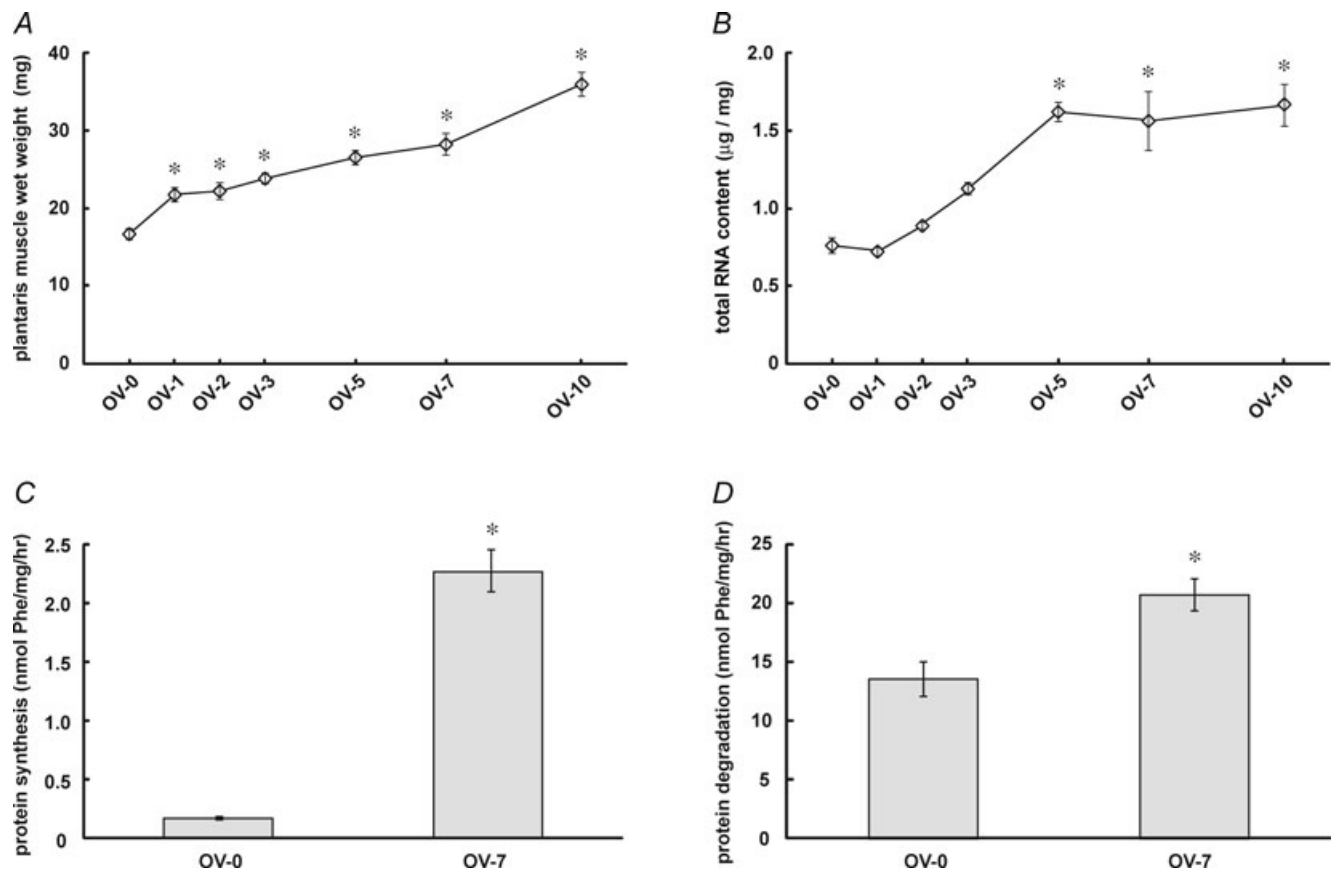


Figure 1. Increased muscle weight, RNA content and protein metabolism in response to mechanical overload of the plantaris muscle

A, plantaris muscle weight (mg) was significantly increased in response to mechanical overload (OV) after 1 day (OV-1) and continued to progressively increase to OV-10 ($n = 4/\text{time point}$). B, total RNA content significantly increased in response to OV after 5 days (OV-5) and remained elevated for the remainder of the time course ($n = 4/\text{time point}$). C, *in vitro* protein synthesis rate showed a significant 14.2-fold increase in the OV-7 group compared to the control group ($n = 8/\text{group}$). D, *in vitro* protein degradation rate was significantly increased 1.7-fold in the OV-7 group compared to the control group ($n = 8/\text{group}$). All results are expressed as the mean $\pm$ SEM with a significant difference ($P < 0.05$) from the control group (OV-0) designated by an asterisk.

both *in vitro* and *in vivo* conditions and this is prevented by treatment with rapamycin, an mTORC1 inhibitor (Brown *et al.* 1995; Brunn *et al.* 1997). Phosphorylation of T421/S424 of S6K1 is suggested to be required for changing the conformation of S6K1 and this allows the sequential phosphorylation of other sites including T389 (Duchene *et al.* 2008). The phosphorylation state of S6K1 at T389 was significantly increased 8.5-fold at day 1 after overload compared to the control group. We also detected a 4.0-fold increase in phosphorylation at the T421/S424

site of S6K1 following 1 day of mechanical overload (OV-1) (Fig. 2A and B). The level of S6K1 phosphorylation peaked between days OV-2 and OV-3 and remained significantly elevated for the entire 10 days of study (Fig. 2B). In addition, to assess the functional significance of activated mTORC1 signalling in skeletal muscle fibre, the phosphorylation state of the downstream effector rpS6 at S235/236 sites was evaluated by immunohistochemistry. Following 1 day of overload, there is a robust increase in the amount of staining for rpS6 S235/236 as compared

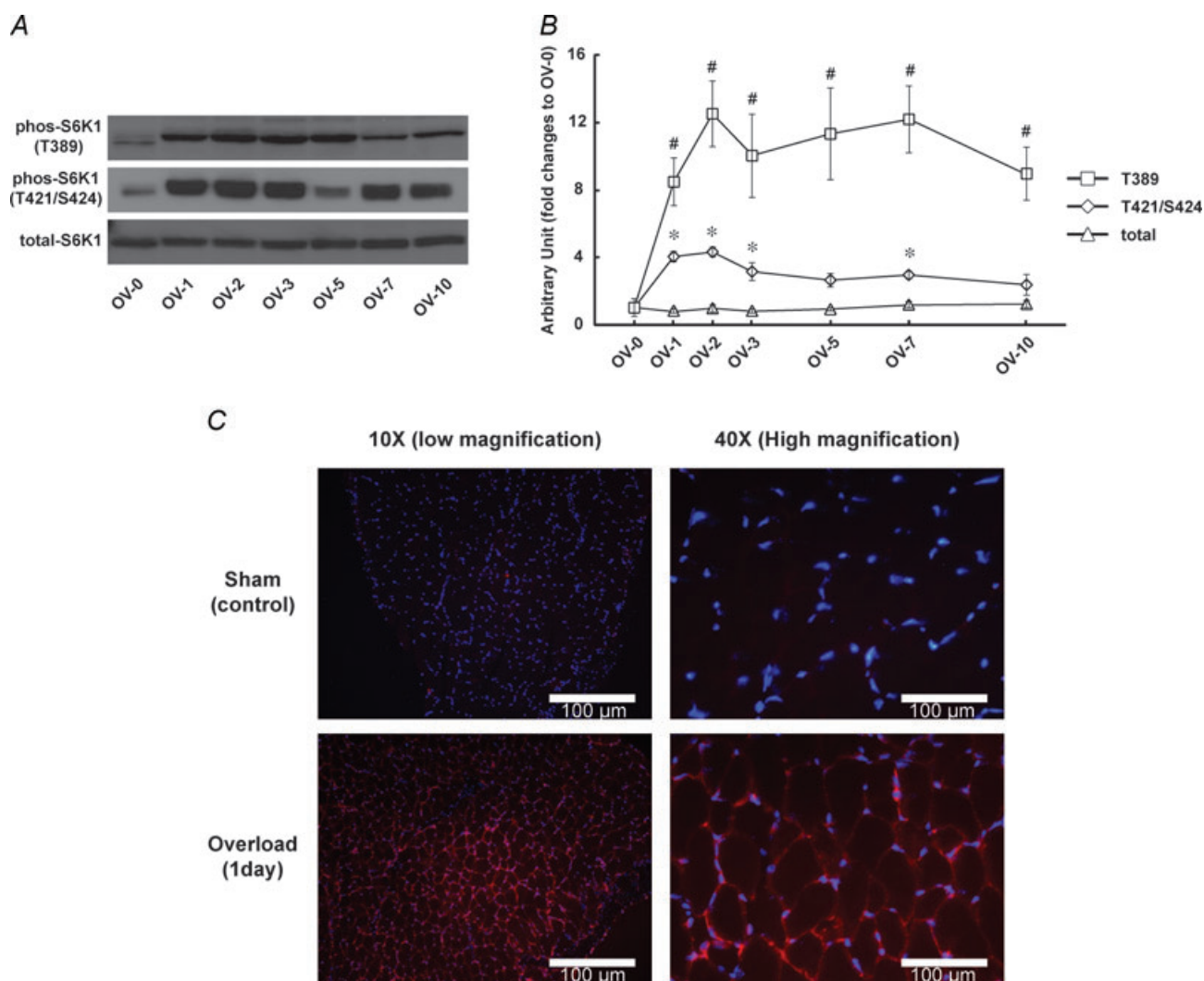


Figure 2. Early activation of mTORC1 signalling in response to mechanical overload

A, representative images of S6K1 showing phosphorylation status of T389 and T421/S424 and total S6K1 protein. B, relative (normalized to OV-0) phosphorylation levels of T389 and T421/S424 sites and total protein level of S6K1 at each time point were quantified ($n = 4-5/\text{group}$). Significant differences: #quantification of S6K1-T389 phosphorylation compared to control (OV-0) group, $P < 0.05$; *quantification of S6K1-T421/S424 phosphorylation compared to control (OV-0) group, $P < 0.05$. All results are expressed as the mean $\pm$ SEM. C, phosphorylation state of rpS6 was determined by immunofluorescence analysis. Upper panels were from sham-operated control muscle and lower panels were from the plantaris muscle with a single day of mechanical overload. rpS6 phosphorylation at S235/236 was detected using Texas Red-conjugated goat anti-rabbit antibody (red), and nuclei were visualized by DAPI (blue). Images with low ($\times 10$, left panels) and high ($\times 40$, right panels) magnification were indicated. Each image was merged using SPOT Advanced Software (SPOT Imaging Solutions, Sterling Heights, MI, USA). Scale bars represent $400\ \mu\text{m}$ in low magnification images and $100\ \mu\text{m}$ in high magnification images.

to the sham-operated control (Fig. 2C). In the plantaris muscle of the sham-operated control, the staining pattern of phosphorylated rpS6 was sparsely distributed at the periphery of the muscle fibre. Following 1 day of mechanical overload, the intensity of immunofluorescent staining for phosphorylated rpS6 was enhanced and was primarily seen within or closely associated with the muscle cell membrane. Localization of phosphorylated rpS6 to the cell membrane is consistent with the localization pattern of ribosomes in skeletal muscle (Horne & Hesketh, 1990). Furthermore, we also examined the localization profile of the endothelial cell marker CD31 to assess the effects of invading inflammatory cells on mTORC1 signalling. It was previously reported that inflammatory cell accumulation, particularly neutrophils, are observed following 1 day of mechanical overload in skeletal muscle (Novak *et al.* 2009). In agreement with Novak *et al.* we do see increased CD31 staining at day 1 following overload, but there is minimal colocalization with phosphorylated rpS6 in the overloaded plantaris muscle (see Fig. 1 in Supplemental material, available online only).

In contrast to the rapid activation of mTORC1 signalling, a significant change in the phosphorylation state of Akt was not detected until OV-2 (S473 site, 1.8-fold) or OV-3 (T308 site, 5.3-fold) following mechanical overload of the plantaris muscle. The change in Akt phosphorylation status was accompanied by a significant increase at OV-2 in the total Akt content (Fig. 3A and B). These results suggest that the early activation of mTORC1 was independent of Akt signalling. To confirm the delayed activation in Akt, we determined the phosphorylation status of direct downstream targets of Akt, including PRAS40 (T246), glycogen synthase kinase 3 β (GSK3- β) (S9) and TSC2 (S939 and T1462). The aforementioned phosphorylation sites of each protein were previously shown to be direct phosphorylation targets of Akt kinase activity (van Weeren *et al.* 1998; Inoki *et al.* 2002; Sancak *et al.* 2007; Huang & Manning, 2008). Phosphorylation of PRAS40 and GSK3- β was not different at OV-1 but it was significantly increased after 3 days of mechanical overload (OV-3) and remained significantly elevated for the duration of the time course with no change in total PRAS40 or GSK3- β levels (Fig. 3C and D, and E and F, respectively). A significant change in TSC2 phosphorylation at S939 or T1462 was not detected until OV-7 and was paralleled by a significant increase in total TSC2 and TSC1 levels. Reciprocal co-immunoprecipitation experiments revealed that the complex formation between TSC1 and TSC2 remained unchanged after 1 and 7 days of mechanical overload compared to control (Fig. 4A–C). These results are consistent with the delayed activation of Akt and support the idea that the early activation of mTORC1 in response to mechanical overload *in vivo* is through an Akt-independent mechanism.

Inhibition of PI3K signalling did not prevent early activation of mTORC1

To determine whether PI3K/Akt-dependent regulation is required for mechanical overload-induced mTORC1 activation, the PI3K inhibitor wortmannin and the mTORC1 inhibitor rapamycin were intraperitoneally administered. To confirm that PI3K/Akt and mTORC1 signalling could be inhibited *in vivo*, we carried out a control experiment using insulin stimulation. Following intraperitoneal injection of insulin, Akt and S6K1 were highly phosphorylated in the plantaris muscle. The PI3K inhibitor wortmannin blocked insulin-induced phosphorylation of Akt and S6K1. In contrast, treatment with the mTORC1 inhibitor, rapamycin, resulted in the complete inhibition of S6K1 phosphorylation, but did not block phosphorylation of the upstream Akt (Fig. 5A). These results confirmed earlier studies showing that growth factor (insulin)-induced regulation of mTORC1 activity is mediated through a PI3K/Akt-dependent signalling pathway (Rommel *et al.* 2001; Miyazaki *et al.* 2010). Next, we investigated the effect of the inhibitors on the mechanical overload-induced mTORC1 activation (Fig. 5B). The mTORC1 activity, as shown by S6K1 T389 phosphorylation, was dramatically increased with mechanical overload after 1 day (OV-1) in the vehicle (DMSO)-treated control group but with no change in phosphorylation state of Akt. Interestingly, inhibition of PI3K signalling with wortmannin did not affect mechanical overload-induced S6K1 phosphorylation at either the T389 or T421/S424 sites. In contrast, administration of the mTOR-specific inhibitor rapamycin completely blocked the S6K1 phosphorylation at T389, and partially prevented S6K1 T421/S424 phosphorylation. Additionally, we observed no change in Akt kinase activity as determined by the phosphorylation state of the GSK3- α peptide following 1 day of mechanical overload, with or without wortmannin or rapamycin treatment (Fig. 5B). These data demonstrate that mTORC1 activation at the early phase of mechanical overload occurs independently of PI3K signalling in skeletal muscle.

Early activation of MEK/ERK signalling in response to mechanical overload

Recent studies have shown that the mitogen-activated protein kinase kinase (MEK)/extracellular signal-regulated kinase (ERK)-dependent pathway can stimulate mTORC1 activity by inhibiting TSC2 activity via phosphorylation at the S664 site (Ma *et al.* 2005, 2007). To determine if a similar mechanism is operative in the plantaris muscle, we examined MEK1/2 and ERK1/2 signalling in response to mechanical overload. As shown in Fig. 6, the relative phosphorylation levels of MEK1/2 (S217/221 site, Fig. 6A and B) and the

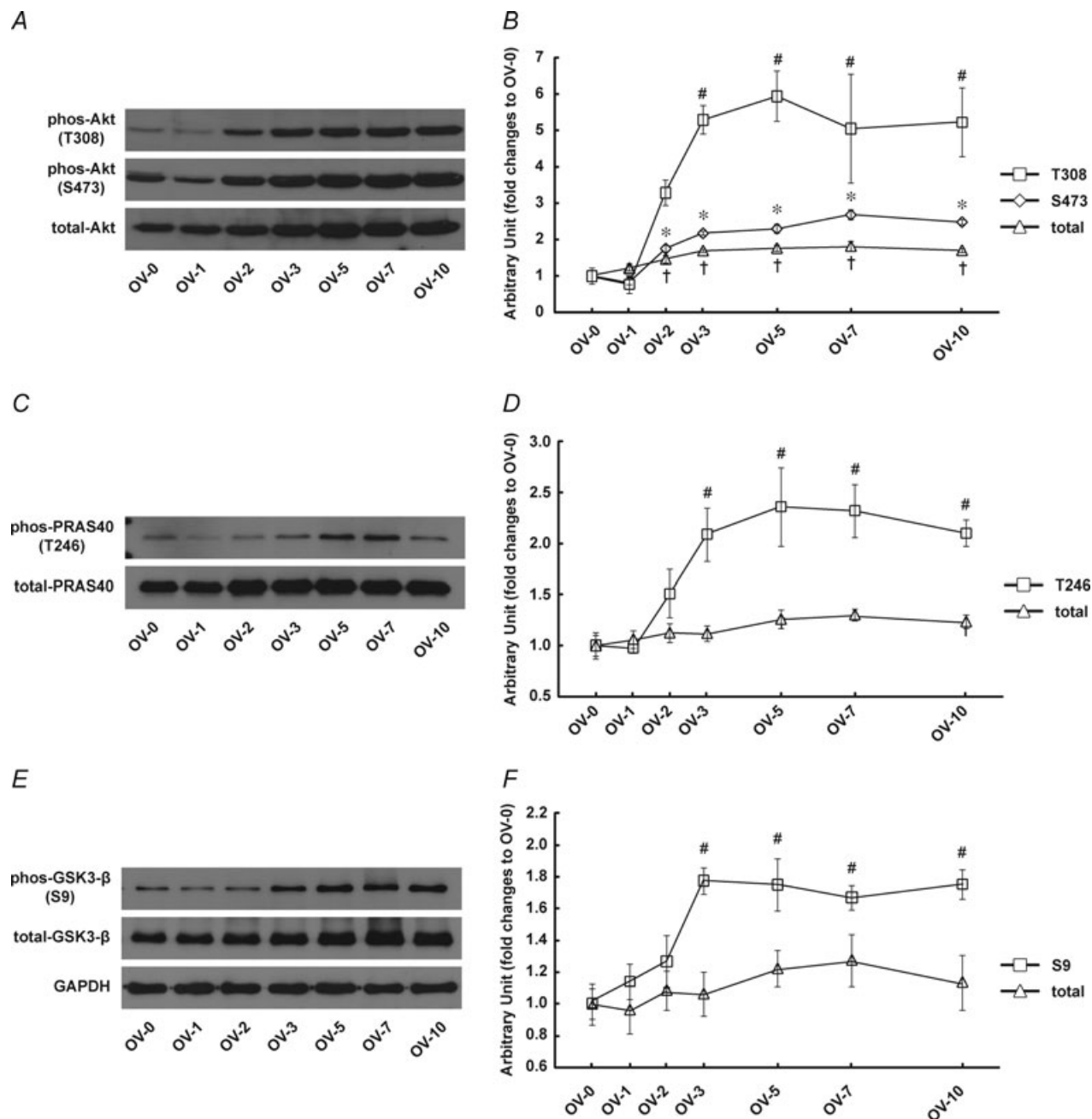


Figure 3. Delayed phosphorylation of Akt and direct Akt targets following mechanical overload of the plantaris muscle

Mechanical overload-induced alteration in the phosphorylation state of Akt and downstream targets of Akt including PRAS40 (at T246 site) and GSK3- β (at S9 site) were determined. *A*, representative images showing Akt phosphorylation (S473 and T308 sites) states and total Akt protein. *B*, relative phosphorylation (S473 and T308 sites) and total protein levels of Akt at each time point were quantified ($n = 4-5/\text{group}$). Significant differences: #quantification of Akt-S473 phosphorylation, compared to the control (OV-0) group, $P < 0.05$; *quantification of Akt-T308 phosphorylation, compared to the control (OV-0) group, $P < 0.05$; †quantification of Akt-total, compared to the control (OV-0) group, $P < 0.05$. *C*, representative images of PRAS40 phosphorylation (T246 site) state and total protein expression. *D*, the relative (normalized to OV-0) phosphorylation (T246 site) and total protein levels of PRAS40 in each group were quantified. Significant differences: #quantification of PRAS40-T246 phosphorylation compared to the control (OV-0) group, $P < 0.05$. *E*, representative images of GSK3- β phosphorylation (S9 site) state and total protein expression. *F*, relative phosphorylation (S9 site) and total protein levels of GSK3- β in each group were quantified. Significant differences: #quantification of GSK3- β -S9 phosphorylation compared to the control (OV-0) group, $P < 0.05$. GAPDH was used as a loading control among the experimental conditions. All results are expressed as the mean $\pm$ SEM ($n = 4-5$).

downstream effector ERK1/2 (T202/204 site, Fig. 6C and D) in the plantaris muscle were significantly increased 3.6-fold and 2.2-fold, respectively, after a single day of mechanical overload (OV-1) compared to the control group (OV-0). MEK1/2 and ERK1/2 phosphorylation remained significantly elevated for at least 7 days after the initiation of mechanical overload (Fig. 6A–D). Furthermore, to assess the functional significance of activated MEK/ERK on mTORC1 activity, we evaluated the phosphorylation state of TSC2 at the S664 site. Following 1 day of overload, there is a clear increase in the amount of TSC2 S664 phosphorylation as compared to the sham-operated control (Fig. 7A). Immunohistochemical staining confirmed the increase in TSC2 phosphorylation following mechanical overload with the majority of the signal being derived from skeletal muscle fibres (Fig. 7B–D). The distinctive pattern of enhanced chromogenic staining was characterized by localization of S664-phosphorylated TSC2 at the muscle cell membrane with higher intracellular levels in individual muscle fibres in overloaded muscle. These observations suggest that

the MEK/ERK-dependent signalling may contribute to mTORC1 activation at the early phase of mechanical overload through phosphorylation of TSC2 at the S664 site.

mTORC1-independent regulation of rpS6 through MEK/ERK/RSK signalling in response to mechanical overload

It has been reported that activation of the mitogen-activated protein kinase (MAPK)-dependent pathway contributes to rpS6 phosphorylation and regulation of translation initiation (Anjum & Blenis, 2008). While S6K1/S6K2 are the main kinases of rpS6, MEK/ERK-dependent activation of p90 ribosomal S6 kinases (RSKs) can regulate rpS6 through phosphorylation at the S235/236 site. This provides for the control of translation initiation machinery through an mTORC1-independent mechanism (Pende *et al.* 2004; Roux *et al.* 2007). To determine whether mTORC1-independent regulation of rpS6 is functional during mechanical overload, we examined RSK signalling

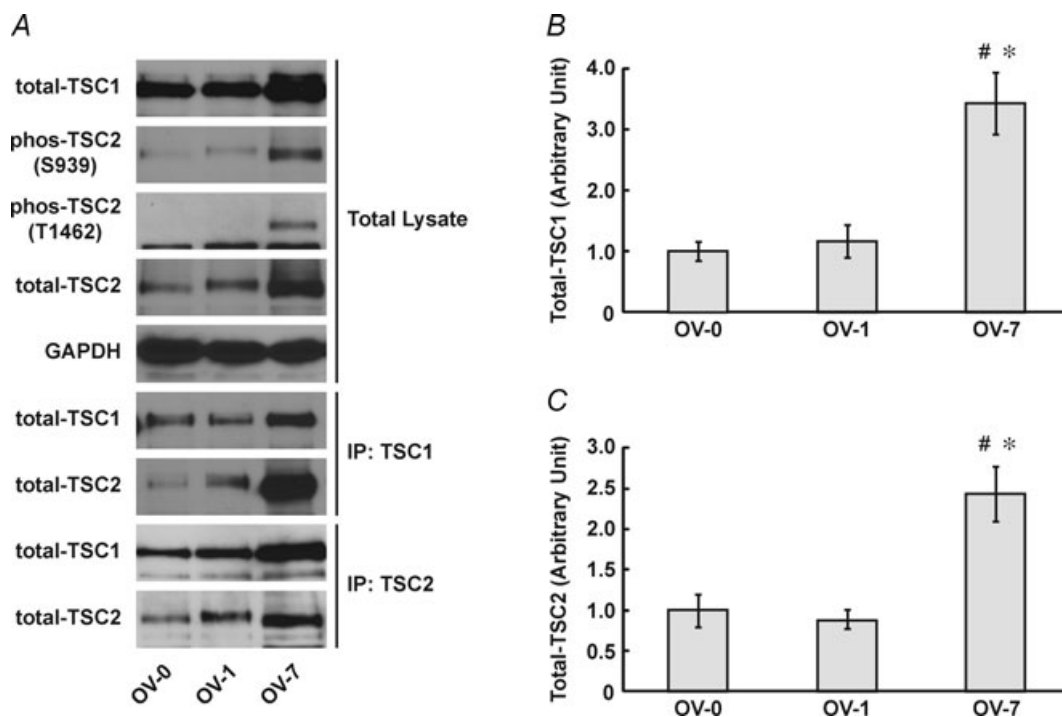


Figure 4. Increased expression of TSC1 and TSC2 following mechanical overload

Mechanical overload-induced phosphorylation of TSC2 (S939 and T1462 sites) and complex formation between TSC1 and TSC2 were determined in the plantaris muscle. A, representative images of phosphorylation (TSC2-S939 and TSC2-T1462) state and total protein expression of TSC1 and TSC2. Total lysates from the plantaris muscle were immunoprecipitated with either anti-TSC1 or anti-TSC2 antibody. GAPDH was used as a loading control among the experimental conditions. B and C, the relative expression (normalized to OV-0) levels of total TSC1 (B) and total TSC2 (C) in each group were quantified. All results are expressed as the mean $\pm$ SEM ($n = 3/\text{group}$). Significant differences: #compared to the control (OV-0) group, *compared to 1 day of mechanical overload (OV-1) group, $P < 0.05$. The phosphorylation level of TSC2 at S939 site was increased in the plantaris muscle following 7 days of mechanical overload with concomitant increase in the total TSC2 protein. Co-immunoprecipitation assay showed that mechanical overload did not affect the physical association between TSC1 and TSC2 in the plantaris muscle. No changes in the phosphorylation state of TSC2 at S939 or T1462 site was observed at day 1 of overload.

toward rpS6 phosphorylation in the plantaris muscle. As shown in Fig. 8, S6K1 at the T389 site and rpS6 at both S235/236 (RSK-dependent residues) and S240/244 (S6K1-dependent residues) sites were highly

phosphorylated following 1 day of mechanical overload. The mTORC1 inhibitor rapamycin completely blocked overload-induced phosphorylation of S6K1. In contrast, rapamycin treatment was not sufficient to achieve a

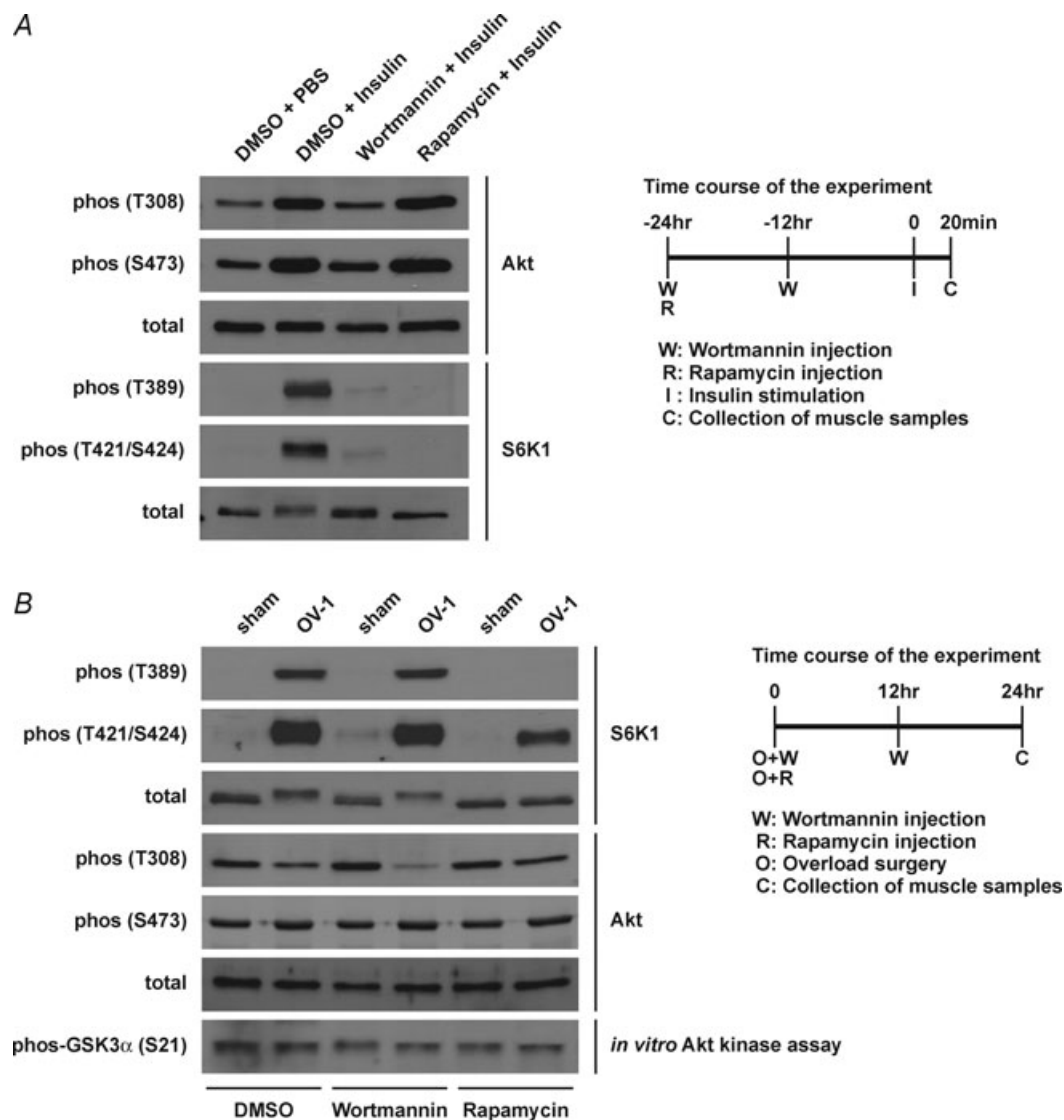


Figure 5. Inhibition of PI3K signalling did not prevent early activation of mTORC1 in response to mechanical overload

A, effect of wortmannin or rapamycin treatment on insulin-induced phosphorylation of Akt (S473 and T308 sites) and S6K1 (T389 and T421/S424 sites) in the plantaris muscle. Insulin (0.75 U kg^{-1}), wortmannin (3 mg kg^{-1}) and rapamycin (1.5 mg kg^{-1}) were administered by intraperitoneal injection. DMSO was used as a solvent for wortmannin and rapamycin. Wortmannin was administered twice (12 h and 24 h prior to the insulin stimulation), and rapamycin was injected once (24 h prior to the insulin stimulation). Twenty-four hours after the first shot of solvent (DMSO) or inhibitors, insulin was administered by intraperitoneal injection and muscle samples collected 20 min after insulin treatment. As expected, wortmannin blocked insulin-induced phosphorylation of Akt and S6K1 whereas rapamycin only prevented S6K1 phosphorylation. **B**, effect of wortmannin or rapamycin treatment on mechanical overload-induced phosphorylation of Akt and S6K1 and *in vitro* Akt kinase activity in the plantaris muscle. Wortmannin was administered twice (immediately and 12 h after the synergist ablation surgery) and rapamycin was injected once (immediately after the surgery). Sham-operated (sham) served as a control. Muscle samples were collected 24 h after administration of DMSO or inhibitor. There was no change in Akt phosphorylation at this early time point (OV-1). However, phosphorylation of S6K1 at T389 was blocked by rapamycin but not by wortmannin. The *in vitro* Akt kinase assay revealed no change in Akt activity confirming the Akt phosphorylation data. Each set of experiments were repeated at least 3 times using different animals.

complete inhibition of rpS6 phosphorylation, as it was still partially phosphorylated at both S235/236 and S240/244 sites in response to mechanical overload. We observed no inhibitory effects of rapamycin treatment on overload-induced activation of the MAPK signalling pathway, as determined by the phosphorylation state of MEK1/2 (S217/221), ERK1/2 (T202/204) and the downstream effector RSK (T359/S363 and S380). These data demonstrate that mTORC1-independent regulation of the MEK/ERK/RSK pathway may contribute to the overload-induced activation of rpS6 phosphorylation in the plantaris muscle.

Discussion

According to the 'somatomedin hypothesis', pituitary growth hormones stimulate whole body growth by

enhancing liver production of somatomedins (IGFs), which in turn stimulate body growth in an endocrine manner (Daughaday & Rotwein, 1989). The contribution of circulating growth hormones and IGF-1, however, to skeletal muscle growth in mature animals has been called into question. It was reported that, in hypophysectomized rats, skeletal muscle hypertrophy occurs to the same extent in the absence of circulating pituitary hormones (Goldberg, 1967). Two independent groups also reported that liver-derived circulating IGF-1 is not required for postnatal body growth (Sjogren *et al.* 1999; Yakar *et al.* 1999). Despite a 75–85% reduction in the concentration of circulating IGF-1, mice showed no significant difference in whole body size, bone length or skeletal muscle mass and were able to fully respond to a hypertrophic stimulus (Matheny *et al.* 2009).

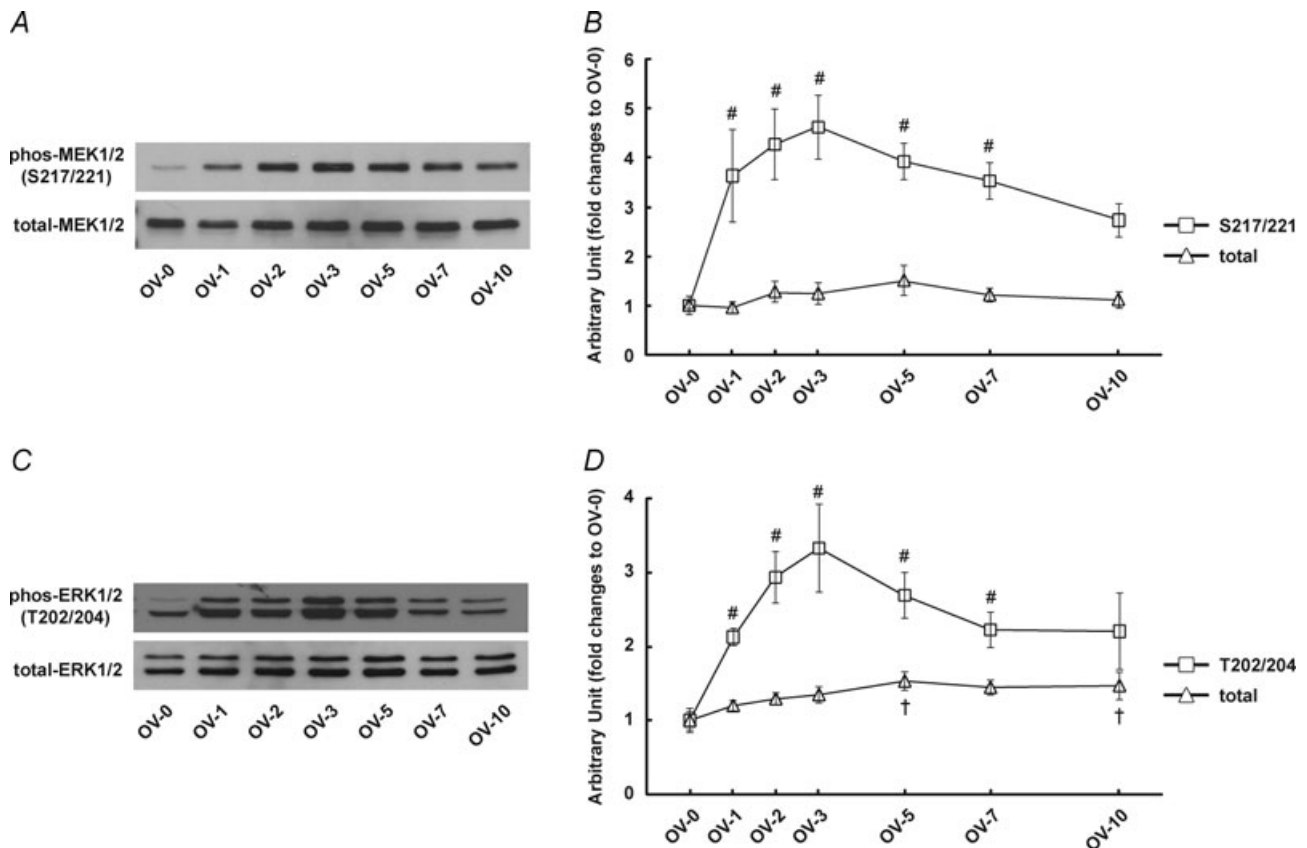


Figure 6. Early activation of MEK1/2 and ERK1/2 signalling in plantaris muscle following mechanical overload

A, representative images of phosphorylation state (S217/221 sites) and total protein expression of MEK1/2. B, relative (normalized to OV-0) phosphorylation (S217/221 sites) and total protein levels of MEK1/2 in each group were quantified. All results are expressed as the mean $\pm$ SEM ($n = 4-5$). Significant differences: #compared to the control (OV-0) group, $P < 0.05$. C, representative images of phosphorylation states (T202/204) and total protein expression of ERK1/2. D, relative (normalized to OV-0) phosphorylation (T202/204) and total protein levels of ERK1/2 in each group were quantified. All results are expressed as the mean $\pm$ SEM ($n = 4-5$). Significant differences: #quantification of ERK1/2-T202/204 phosphorylation compared to the control (OV-0) group, $P < 0.05$; †quantification of ERK1/2-total compared to the control (OV-0) group, $P < 0.05$. For both MEK1/2 and ERK1/2 there was a significant increase in phosphorylation after 1 day of overload (OV-1) that was maintained until OV-7.

IGF-1 is also synthesized locally in extra-hepatic tissues including skeletal muscle, brain and kidney where it acts in an autocrine/paracrine manner (Isgaard *et al.* 1989). The local production of IGF-1 within the muscle has been thought to be responsible for the induction of skeletal muscle hypertrophy, especially in response to increased workload (DeVol *et al.* 1990; Hambrecht *et al.* 2005). This IGF-1-dependent hypertrophy of skeletal muscle is mediated through ligand binding to the tyrosine kinase IGF-1 receptor on the muscle cell membrane with the subsequent activation of downstream intracellular signalling events, primarily through the PI3K/Akt-dependent pathway (Bodine *et al.* 2001; Rommel *et al.* 2001). Thus, it has largely been assumed that the local production of IGF-1 in skeletal muscle was responsible for the increased activation of the PI3K/Akt signalling cascade during mechanical overload, leading in turn to the stimulation of mTORC1 activity and enhanced protein synthesis (Glass, 2005). In contrast to this scenario, previous reports have found a poor correlation between the mechanical load-induced increase in IGF-1 mRNA or protein and activation of mTORC1-dependent signalling and subsequent increase in protein synthesis rate (Spangenburg, 2009). Adams and coworkers have shown that overload-induced IGF-1 production in skeletal muscle occurred between 24 and 72 h after the initiation of the workload (Adams & Haddad, 1996; Adams *et al.* 1999), while activation of

mTORC1-dependent signalling was induced within a few hours of the onset of mechanical stimulation (Baar & Esser, 1999; Nader & Esser, 2001). Together, these observations support the possibility that mechanical load-induced activation of mTORC1 signalling is independent of the IGF-1 signalling pathway.

In support of this possibility, studies using a muscle-specific dominant-negative model of the IGF-1 receptor showed that a functional IGF-1 receptor-dependent mechanism was not necessary for mTORC1 activation and the induction of skeletal muscle hypertrophy in response to increased mechanical load (Spangenburg *et al.* 2008) or a single bout of high frequency muscle contractions (Witkowski *et al.* 2010). Furthermore, it was shown that the PI3K/Akt-dependent regulation was not required for the activation of mTORC1 signalling following eccentric contractions in skeletal muscle under *ex vivo* conditions (O'Neil *et al.* 2009). A more recent report also showed that over-expression of Ras homologue enriched in brain (Rheb), which is a direct activator of mTORC1, was sufficient to induce skeletal muscle hypertrophy independent of PI3K/Akt signalling (Goodman *et al.* 2010). The data reported here clearly support these earlier observations and the idea that the initial activation of mTORC1 in response to mechanical overload does not involve IGF-1/PI3K/Akt-dependent signalling. Consistent with earlier reports, we found that the early activation of mTORC1 signalling (within

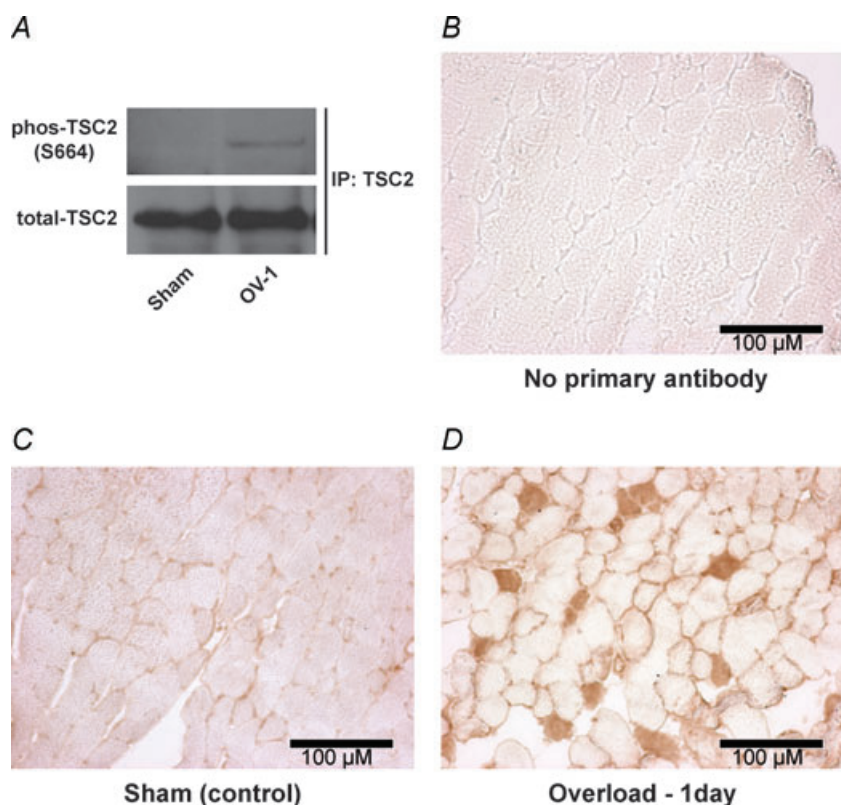


Figure 7. Increased TSC2 phosphorylation following 1 day of mechanical overload

Phosphorylation state of TSC2 at S664 site was examined by Western blotting and chromogenic immunohistochemical analysis. A, representative images of TSC2 phosphorylation at S664 site and total protein expression of TSC2. Total lysates from the plantaris muscle were immunoprecipitated with anti-TSC2 antibody. For the Western blotting analysis, phospho-specific antibody for TSC2 at the S664 site was kindly provided by Dr Pier Paolo Pandolfi, Beth Israel Deaconess Medical Center, Harvard Medical School. B–D, the plantaris muscles for immunohistochemical analysis were collected from sham-operated control (Sham) and 1 day of mechanical overload. For immunohistochemistry, the phospho-specific antibody for TSC2 at S664 site was obtained from Biolegend. The absence of staining in the no antibody control (B) demonstrated no background staining whereas the sham control (C) showed very low TSC2 phosphorylation under resting conditions. Following a single day of overload (D) muscle fibres showed enhanced chromogenic staining indicating increased TSC2 phosphorylation at site S664. Scale bars represent 100 μm in all images.

1 day following the onset of mechanical overload) occurred independently of PI3K/Akt regulation in skeletal muscle *in vivo*. Specifically, there was no change in Akt phosphorylation or kinase activity as well as no alteration in the phosphorylation of direct downstream targets of Akt after 1 day of mechanical overload of the plantaris muscle. It is important to note, however, that inhibition of the early activation of mTORC1-dependent signalling does not block the subsequent hypertrophic growth of skeletal muscle *in vivo*. We confirmed that *in vivo* treatment with either wortmannin or rapamycin (up to the first 24 h after surgery) did not prevent compensatory hypertrophy

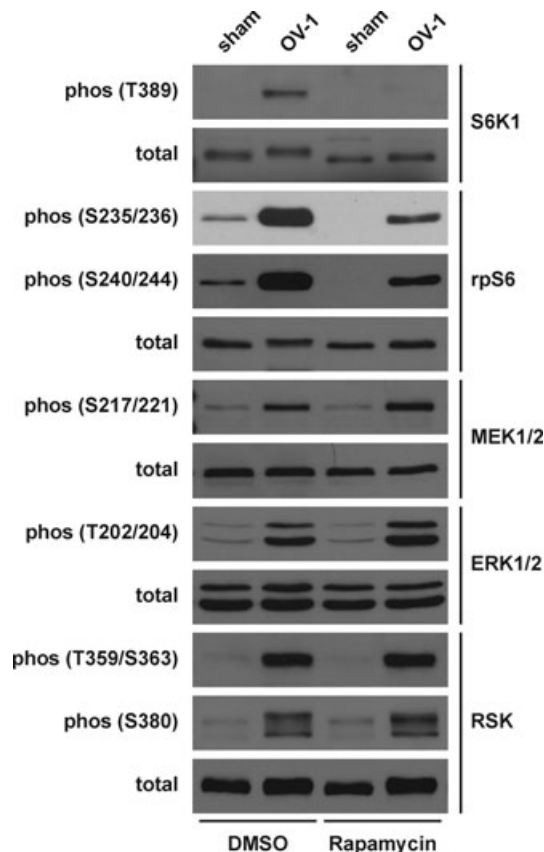


Figure 8. mTORC1 inhibitor rapamycin did not prevent MEK/ERK/RSK signalling in response to mechanical overload

The effect of rapamycin treatment on mechanical overload-induced phosphorylation of S6K1 (T389), rpS6 (S235/236 and S240/244), MEK1/2 (S217/221), ERK1/2 (T202/204) and RSK (T359/S363 and S380) was examined. Rapamycin (1.5 mg kg^{-1}) was administered immediately after surgery by intraperitoneal injection. DMSO was used as a solvent for rapamycin. Sham-operated (sham) served as a control. Muscle samples were collected 24 h after administration of DMSO or rapamycin. The mTORC1 inhibitor rapamycin completely blocked overload-induced phosphorylation of S6K1; however, it was not sufficient for complete inhibition of rpS6 phosphorylation at both S235/236 and S240/244 sites in response to mechanical overload. No inhibitory effects of rapamycin treatment on overload-induced phosphorylation of MEK1/2, ERK1/2 and RSK were observed. Each set of experiments was repeated at least 3 times using different animals.

in the plantaris muscle following 7 days of mechanical overload. Additionally, we found no inhibitory effects of the single-day treatment on the phosphorylation state of Akt and S6K1 at 7 days following overload (Supplemental Fig. 2). These data suggest that the long term activation of Akt-dependent or mTOR-dependent signalling following mechanical overload is not dependent on specific or unique events that occur on day 1.

It has been reported that MEK/ERK signalling and its downstream effectors are rapidly phosphorylated and activated following anabolic stimuli such as resistance exercises in skeletal muscle (Drummond *et al.* 2009). The results from the current study represent the first evidence in skeletal muscle suggesting MEK/ERK signalling contributes to activation of mTORC1 signalling through phosphorylation of TSC2 at the S664 site (Fig. 9). Together with TSC1, TSC2 inhibits mTORC1 activity through its function as a GTPase-activating protein (GAP) for a small G protein Rheb (Inoki *et al.* 2003). Phosphorylation of TSC2 at both the S664 and the S540 sites has been shown to be directly phosphorylated by ERK1/2 and contributes to enhanced mTORC1 activity, probably through inhibiting TSC2 GAP activity thereby

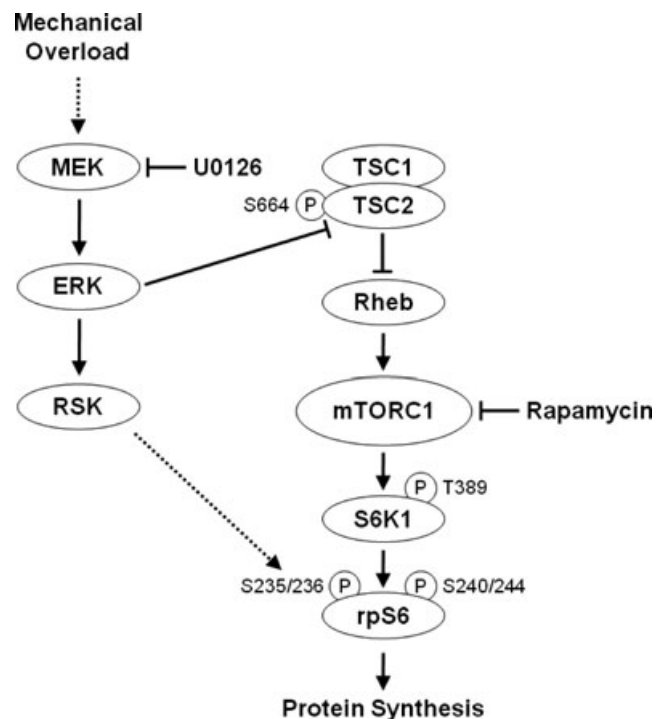


Figure 9. Schematic representation of the molecular mechanisms in the regulation of mTORC1 signalling pathway during early phase of mechanical overload-induced skeletal muscle hypertrophy

Activation of mTORC1 signalling at the early phase in response to mechanical overload occurs independently of PI3K/Akt signalling in skeletal muscle. MEK/ERK pathway through TSC2 phosphorylation at S664 may contribute to mTORC1 activation during skeletal muscle hypertrophy.

allowing Rheb to accumulate in its active GTP-bound form (Ballif *et al.* 2005; Ma *et al.* 2005, 2007). Unfortunately, *in vivo* administration of the MEK inhibitors U0126 or AZD6244 (U0126, 200 mg kg⁻¹; AZD6244, 100 mg kg⁻¹) did not completely prevent ERK1/2 phosphorylation in skeletal muscle and higher doses were not healthy for the mice (data not shown). A genetic approach employing the muscle-specific inactivation of ERK1/2 will be required to investigate more mechanistically the MEK/ERK-dependent pathway in the regulation of mTORC1 activity in skeletal muscle. In addition to our focus on mTORC1 function, our results suggest that the MEK/ERK/RSK pathway may contribute to mechanical overload-induced activation of rpS6 phosphorylation independently of mTORC1 (Fig. 9). It is important to note, however, that groups have shown that treatment with the mTORC1-inhibitor rapamycin blocks changes in rates of protein synthesis in skeletal muscle following an increased level of contractile workload both in rodents (Kubica *et al.* 2005) and humans (Drummond *et al.* 2009). Thus, the mTORC1-independent mechanisms (e.g. MEK/ERK/RSK pathway) may not be sufficient for increases in protein synthesis in skeletal muscle and they may play a role in the magnitude or duration of increases in protein synthesis following resistance exercise/mechanical overload.

Summary

We determined the temporal activation of mTORC1-dependent signalling by PI3K/Akt and MEK/ERK pathways in response to a hypertrophic stimulus. The primary findings of the study were: (1) mechanical overload of the plantaris muscle resulted in robust activation of Akt, MEK/ERK and mTORC1-dependent pathways over 10 days; (2) mTORC1 activation, as determined by phosphorylation of S6K1, was detected after 1 day of mechanical overload which was prior to increased Akt-dependent signalling; (3) treatment with PI3K inhibitor wortmannin was sufficient to inhibit growth factor-dependent signalling but did not prevent overload-induced mTORC1 activation *in vivo*; and (4) MEK/ERK-dependent signalling may contribute to mTORC1 activation at the early phase of mechanical overload through phosphorylation of TSC2 at the S664 site. These findings provide strong evidence that the early activation of mTORC1 signalling in response to mechanical overload occurs independently of PI3K/Akt signalling. Further, we suggest that MEK/ERK inhibition of TSC2 contributes to mTORC1 activation during skeletal muscle hypertrophy.

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Author contributions

M.M.: conception, design, analysis, interpretation of data and writing; J.J.M.: conception, design, interpretation of data and writing; M.J.E.: analysis, interpretation of data and writing; K.A.E.: conception, design, interpretation of data and writing. Experiments on the measurement of protein synthesis/degradation were carried out in the School of Kinesiology, University of Illinois at Chicago by M.J.E. All other analyses in this study were carried out in the Department of Physiology and Center for Muscle Biology, University of Kentucky by M.M.

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